Steps for using and calibration application areas of the AILEIKE ALKB202 portable micro-pressure pump

The AILEIKE ALKB202 portable micro-pressure pump is a hand-operated micro-pressure source that is easy and effortless to operate. It features a smooth pressure adjustment process, excellent sealing, and simple daily maintenance. It can generate both positive and negative pressures, making it an indispensable device for micro-pressure instrument calibration in laboratory precision verification and industrial on-site mobile rapid calibration. The following will provide a detailed introduction from five dimensions: technical specifications, operation methods, application scenarios, usage precautions, and product summary.

1. Technical indicators

name	Lightweight micro-pressure pump	Pressurization method	manual
model	ALKB202	medium	clean air
pressure range	-90~600kPa	Overall dimensions	220×90×80mm
Adjust fineness	10 Pa	weight	1.8kg
Output port	2 pieces	Pressure connection	Hose connector thread (M20*1.5)

II. Operation method

First, place the lightweight micro-pressure pump on a flat workbench (in a non-corrosive, vibration-free, and dust-free environment).

Install the standard gauge and the verified micro pressure instrument (pressure transmitter, micro pressure gauge, pressure sensor, etc.).

(I) Pressure gauge calibration process

1. Voltage boosting process

First, rotate the fine adjustment handle counterclockwise until it stops, and then close the vent valve. Use the pressure adjustment handle to rotate clockwise to increase pressure, and pressurize to the verification value.

2. Decompression process

Use the pressure regulating handle to rotate counterclockwise to reduce the pressure to the verification value.

3. Pressure relief process

Open the vent valve, gradually release the pressure inside the pipeline until it reaches 0, and then the measurement is complete.

(II) Pressure gauge vacuum calibration process

1. Voltage boosting process

Adjust the pressure regulating handle clockwise until it stops, and then close the vent valve. Use the pressure

regulating handle to slowly adjust counterclockwise to the verification value.

2. Decompression process

Use the pressure regulating handle to rotate clockwise to reduce the pressure, until it reaches the verification value.

3. Pressure relief process

Open the vent valve, gradually release the pressure inside the pipeline until it reaches 0, and then the measurement is complete.

III. Application Scenarios

Laboratory metrological verification:

Calibration standards for metrology institutions are used for periodic calibration and accuracy verification of precision instruments such as micro-pressure gauges, micro-pressure sensors, and pressure switches.

Operation and maintenance of industrial field equipment:

Applied to industrial scenarios such as chemical, power, and metallurgical industries, for daily inspection, troubleshooting, and on-site calibration of pressure instruments.

Scientific research and teaching experiment:

It is suitable for pressure detection experiments, pressure parameter adjustments, and other scientific research and teaching activities in universities and research institutes. With simple operation and stable pressure, it can meet the accuracy requirements of various micro-pressure experiments.

Instrument production and debugging:

Factory testing and aging sampling inspection for micro-pressure gauge, micro-pressure sensor, and transmitter production workshop.

IV. Precautions

1. The ALKB202 portable micro-pressure pump should be used to increase and decrease pressure slowly to avoid pressure shock.

2. After the operation is completed, the pressure must be released to zero before the inspected instrument can be removed.

3. Regularly inspect the sealing rings and joint tightness, promptly replace aging and damaged parts, and ensure the airtightness and calibration accuracy of the equipment.

4. If not in use for an extended period, the equipment should be stored in a dry environment free from corrosive gases.

5. Avoid using it in environments with highly corrosive gases and dust particles.

V. Conclusion

The AILEIKE ALKB202 portable micro-pressure pump features lightweight design, positive and negative pressure ranges, and stable pressure output, addressing the shortcomings of traditional micro-pressure calibration instruments, such as complex operation and limited pressure control accuracy. With its simple structure and easy daily maintenance, it meets both laboratory precision verification and outdoor industrial field mobile calibration needs, facilitating micro-pressure instrument calibration, testing, and debugging

across various industries. It improves the efficiency of micro-pressure detection and is a practical device in the field of micro-pressure calibration.